

A closed-form implied-volatility correction at a cubic local-volatility knot pinned at the money

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Abstract

The standard practice is to parametrise the *local* volatility σ_{loc} and push it to implied volatility through the leading-order Berestycki–Busca–Florent (BBF) inverse-harmonic-mean map. BBF can calibrate essentially any market smile; its weakness is not fit but consistency — being only leading order, the implied vol it returns is not the model’s true Dupire implied vol, so a surface that matches quotes can still carry small static-arbitrage inconsistencies. The Henry-Labordère $\sigma_1 T$ correction (PHL1) is the next order and much closer to the true Dupire implied vol; moving from BBF to PHL1 buys *arbitrage reassurance*, not extra calibration reach. But PHL1 on a single smooth cubic is rigid, and admitting a knot — the natural way to recover flexibility — breaks it: a piecewise-cubic σ_{loc} is only C^2 at its knots (the cubic coefficient $\gamma = \sigma_{\text{loc}}'''$ jumps), while the σ_1 expansion is derived assuming C^3 , leaving a localised residual at the knot. We close that residual in the practically relevant case — a knot pinned exactly at-the-money — in closed form: at first order in the coefficient jump it is a one-dimensional Brownian-bridge integral $\Phi_{BB}(x, 0)$ with peak $3\sqrt{2\pi}/128$ and no fitted constants. This makes “piecewise-cubic with an at-the-money knot, mapped through PHL1” a usable parametrisation that keeps PHL1’s arbitrage reassurance while regaining calibration flexibility. The at-the-money restriction is essential, not cosmetic: off-the-money the same construction needs an unbounded-polynomial truncation, which the $w = 0$ case removes. We validate against the Dupire PDE on parameters calibrated from real SPXW surfaces.

1 Introduction

Calibrating a volatility surface, practitioners fit a parametric *local* volatility (for the monotonicity and regularity an optimiser needs) and map it to implied volatility with the leading-order BBF inverse-harmonic-mean formula. BBF is flexible enough to calibrate essentially any smile, so the motivation for going further is not fit. It is *arbitrage reassurance*: BBF is only leading order, so the implied vol it returns is not the local-vol model’s true Dupire implied vol, and a quote-matching surface can still violate static no-arbitrage (butterfly, calendar) at the sub-basis-point level that an arbitrage-sensitive consumer cannot tolerate. The next-order PHL1 map is much closer to the true Dupire implied vol and removes most of that inconsistency.

The price of PHL1 is rigidity: a single smooth cubic mapped through PHL1 spans far fewer smile shapes than the loose BBF fit. The natural fix — a piecewise-cubic σ_{loc} with a knot — fails, because PHL1’s σ_1 heat-kernel term is derived for C^3 surfaces whereas a piecewise cubic is only C^2 at a knot; the result is a localised implied-vol residual at the knot strike (BBF, a harmonic-mean integral, is insensitive to it). This paper removes that residual in closed form for a knot pinned at-the-money, so the combination “piecewise-cubic-with-ATM-knot + PHL1” becomes usable: it keeps

PHL1’s arbitrage reassurance while recovering the calibration flexibility a smooth cubic lacks. The correction is a parameter-free Brownian-bridge kernel; the at-the-money restriction is what keeps it bounded. The same derivation answers, for the C^2 case in log-normal Black IV, the open question Costeanu and Pirjol [Costeanu and Pirjol, 2011] left in their §5.25 after settling the C^0 , normal-IV case. The remainder derives the correction and validates it against the Dupire forward PDE [Dupire, 1994].

2 Related work

The map from a parametric local-vol surface to implied vol through a small-time asymptotic is one strand of a wider literature on parametric volatility surfaces. The direct predecessor for the knot-residual question, the broader asymptotic-map literature, and the orthogonal parametric and arbitrage-framing literatures are taken in turn.

Direct prior art and open question. The closest predecessor is Costeanu–Pirjol [Costeanu and Pirjol, 2011], who set up the same Duhamel-perturbation framework for the analogous problem one regularity class harsher: a C^0 local vol with a jump in σ'_{loc} at the forward, in *normal* (Bachelier) implied vol. They prove that the ATM normal IV picks up a non-analytic $\sqrt{T}$ term with universal coefficient $\frac{1}{16}\sqrt{\pi/2} \approx 0.0784$ proportional to the jump, and their Table 5.1 lists the powers of T that can arise from each order of perturbation in the same framework. Their §5.25 leaves explicitly open whether higher powers of T ($T^{3/2}$, $T^{5/2}$, ...) actually appear for smoother knot regularities. The contribution here answers that question for the C^2 case (the natural smoothness of a piecewise-cubic calibrator, where only $\sigma'''_{\text{loc}} = \gamma$ jumps) in *log-normal* Black IV: the corresponding entry of their Table 5.1 is the $T^{3/2}$ slot (row $n = 1$, one column past the $\sqrt{T}$ slot they filled), and its universal coefficient is $3\sqrt{2\pi}/128 \approx 0.0588$ — the C^2 companion to their C^0 constant within the same Duhamel lineage. The kernel is the one-dimensional Brownian-bridge integral $\Phi_{BB}(x, 0)$ derived in Section 3 below.

Higher-order LV-to-IV asymptotics. The PHL1/GHLOW2 line derives from the heat-kernel expansion of the call-price PDE: BBF [Berestycki et al., 2002] is leading order, Henry-Labordère [Henry-Labordère, 2008] adds σ_1 , and Gatheral, Hsu, Laurence, Ouyang and Wang [Gatheral et al., 2012] add σ_2 . An alternative methodology, due to Pagliarani–Pascucci [Pagliarani and Pascucci, 2012] and extended by Lorig–Pagliarani–Pascucci [Lorig et al., 2017], builds explicit implied-vol expansions from an adjoint formulation of the pricing PDE; it covers local-stochastic models and produces series without special functions. Its main scheme is a Taylor expansion of the coefficients around a chosen expansion point under the explicit hypothesis $\sigma \in C^N$ there — exactly the assumption a knot at the expansion point violates — so the knot case considered here lies outside its stated hypotheses and is not subsumed by it. De Marco’s harmonic-mean characterisation [De Marco, 2021] pins what the leading-order map is approximating, and hence what the next-order gap (the object PHL1 plus the correction close) actually is.

Direct implied-vol parametrisations. The other side of the design choice is to parametrise the implied-vol surface directly. SABR [Hagan et al., 2002] fits a leading-order asymptotic of a stochastic-vol model. SVI [Gatheral, 2004] and its arbitrage-free refinement SSVI [Gatheral and Jacquier, 2014] drop the dynamic model and write the smile itself as a parametric form, then constrain the parameters to enforce static no-arbitrage. Andreasen–Huge [Andreasen and Huge, 2011] fits a piecewise-constant local vol in one implicit-Euler step of the Dupire PDE; the resulting

prices are arbitrage-free in the sense of the discretised scheme, with as many vol pieces as quotes. This is a different design point from ours: the calibration carries a numerical scheme inside it, there is no closed-form $\sigma_{IV}(k; \theta)$ to feed an optimiser, and “arbitrage-free” is the one-step finite-difference notion rather than the continuous Dupire model’s. The parametric-LV-through-PHL1 route trades that for a closed-form asymptotic on the local-vol side, with arbitrage reassurance carried by PHL1’s tightness against the true Dupire implied vol.

Static arbitrage and asymptotic constraints. “Arbitrage reassurance” above is what Roper [Roper, 2010] catalogues as static no-arbitrage on the IV surface (butterfly, calendar). At the opposite end of the strike axis, Lee’s moment formula [Lee, 2004] gives the universal asymptotic bound on the deep-wing slope of any arbitrage-free IV smile, and thereby on every map in the family this paper works in. The contribution sits between the two: at ATM, where Lee’s tail bound is silent and the static-arbitrage conditions reduce to a single local statement at $k = 0$ – and where the C^3 -regularity failure of a piecewise-cubic local vol is what PHL1 sees.

3 Setup

Work in log-moneyness $k = \log(K/F)$, forward measure ($F = 1$, $r = 0$), single maturity. Local volatility is one cubic, in annualised % units,

$$\sigma_{\text{loc}}(k) = \sigma + \beta k + \alpha k^2 + \gamma k^3. \quad (1)$$

A knot at the money adds a one-sided jump δ in the cubic coefficient,

$$\sigma_{\text{loc}}(k) = \sigma + \beta k + \alpha k^2 + \gamma k^3 + \delta k^3 H(k), \quad (2)$$

with H the Heaviside step. Because k^3 and its first two derivatives vanish at $k = 0$, (2) is C^2 at the knot; only σ_{loc}''' jumps by δ . This is precisely the structure a piecewise-cubic calibrator produces, with the knot moved to ATM.

We use the Black $T = 1$ convention; scale = $\sqrt{252/n} \cdot 100$ maps total vol to annualised % for $n = \max(1, \text{DTE})$ business days, and $\sigma_{\text{total}} = \sigma/\text{scale}$ is the unperturbed ATM total vol.

The asymptotic maps. The leading-order BBF map [Berestycki et al., 2002] is the inverse harmonic mean

$$\text{iv}_{\text{hm}}(k) = k / \int_0^k \frac{dy}{\sigma_{\text{loc}}(y)}, \quad \text{iv}_{\text{hm}}(0) = \sigma_{\text{loc}}(0), \quad (3)$$

whose use as a parametric local-vol-to-implied-vol representation, and its structure, are studied in detail by De Marco [De Marco, 2021]. The next-to-leading heat-kernel expansion of Henry-Labordère [Henry-Labordère, 2008] adds a $\sigma_1 \cdot T$ term, giving $\sigma_{IV} \approx \text{iv}_{\text{hm}} + \sigma_1$. We use the explicit per-strike closed form of that correction as stated by Gatheral, Hsu, Laurence, Ouyang and Wang [Gatheral et al., 2012, Thm. 2.4], and keep the conventional label **PHL1** for it:

$$\sigma_1(k) = \frac{\text{iv}_{\text{hm}}(k)^3}{2k^2} \log \frac{\sqrt{\sigma_{\text{loc}}(0) \sigma_{\text{loc}}(k)}}{\text{iv}_{\text{hm}}(k)}. \quad (4)$$

(Henry-Labordère’s own local-to-implied first-order result is the same correction in the equivalent Hagan–Woodward midpoint form.) The GHLOW2 refinement adds a further $\sigma_2 T^2$ term from the same source [Gatheral et al., 2012]. BBF0 is a harmonic mean of σ_{loc} : a C^2 knot only jumps σ_{loc}''' ,

so the integrand $1/\sigma_{\text{loc}}$ stays C^2 and iv_{hm} is smooth across the knot — BBF0 has no knot-specific error, only the generic leading-order gap present everywhere. The σ_1 expansion underlying PHL1 (and hence GHLOW2) is, by contrast, derived assuming $\sigma_{\text{loc}} \in C^3$; at a C^2 -only knot it leaves a localised residual — exactly the term this note supplies.

4 The at-the-money knot correction

The implied-vol response to (2) expands in the jump δ about the knotless case. At first order, in dimensionless strike $x = k/\sigma_{\text{total}}$,

$$\delta\sigma_{\text{IV}}(k) = \delta \cdot \sigma_{\text{total}}^3 \cdot \Phi_{BB}(x, 0) + O(\delta^2). \quad (5)$$

Very simple derivation. First order in δ , the source is $v(y) = 2\sigma\delta y^3 H(y)$. Inserting it once into the Duhamel expansion of the Dupire call price against the constant- σ Black heat kernel, the Black strike-gamma identity cancels the explicit σ , and a Gaussian completion turns the spatial integral into an expectation under a Brownian bridge pinned at $(0, 0)$ and (k, T) : writing $\lambda = t/T$, the bridge has $y \sim \mathcal{N}(\lambda k, \sigma_{\text{total}}^2 \lambda(1 - \lambda))$. The remaining y -integral against $y^3 H(y)$ is a truncated third moment of a normal,

$$f(\eta) = (\eta^3 + 3\eta) \Phi(\eta) + (\eta^2 + 2) \phi(\eta), \quad (6)$$

with Φ, ϕ the standard normal cdf/pdf. The $\sqrt{\lambda}$ from the Gaussian completion cancels the $1/\sqrt{t}$ of the strike-gamma, the outer time integral collapses to a clean bridge-time integral, and dividing by the Black vega gives (5) with

$$\Phi_{BB}(x, 0) = \int_0^1 (\lambda(1 - \lambda))^{3/2} f(\eta(\lambda; x)) d\lambda, \quad \eta(\lambda; x) = x\sqrt{\frac{\lambda}{1 - \lambda}}. \quad (7)$$

At the knot centre $\eta \equiv 0$, $f(0) = 2/\sqrt{2\pi}$, and the integral is a Beta function:

$$\Phi_{BB}(0, 0) = \frac{2}{\sqrt{2\pi}} B\left(\frac{5}{2}, \frac{5}{2}\right) = \frac{3\sqrt{2\pi}}{128} \approx 0.05875. \quad (8)$$

Why at-the-money. The general knot at offset $w = k_{\text{knot}}/\sigma_{\text{total}}$ keeps a factor $\eta = [\lambda x - (1 - \lambda)w]/\sqrt{\lambda(1 - \lambda)}$; the subtraction below then leaves a polynomial that grows without bound and needs an ad-hoc truncation to stay finite. Setting $w = 0$ removes the $(1 - \lambda)w$ term entirely and the correction is bounded with no hack — the reason this note restricts to the ATM knot.

Avoiding double counting. PHL1 already moves under (2) through its own iv_{hm} and σ_1 . Taylor-expanding (4) in δ at $w = 0$ gives, for $x > 0$, iv_{hm} varies as $x^3/4$ and σ_1 as $x/4$ (both zero for $x \leq 0$). Subtracting these so they are not counted twice yields the *directed* kernel, which decays on both sides:

$$\Phi_{BB}^{\text{dir}}(x, 0) = \begin{cases} \Phi_{BB}(x, 0) - \frac{x^3}{4} - \frac{x}{4}, & x > 0, \\ \Phi_{BB}(x, 0), & x \leq 0. \end{cases} \quad (9)$$

The corrected method, with PHL1 evaluated on the perturbed surface so the cancellation is exact to first order, is

$$\sigma_{\text{IV}}^{\text{PHL1+corr}}(k) = \sigma_{\text{IV}}^{\text{PHL1}}(k) + \delta \cdot \sigma_{\text{total}}^3 \cdot \Phi_{BB}^{\text{dir}}(k/\sigma_{\text{total}}, 0). \quad (10)$$

The bridge integral (7) is evaluated with a 32-point Gauss–Legendre rule.

5 Results

Parameters come from real SPXW surfaces calibrated with a 20-parameter piecewise-cubic local-vol model; we keep the centre cubic and (for the knot case) the first-knot coefficient jump moved to ATM. The Dupire forward PDE (Rannacher start then Crank–Nicolson) is the ground truth.

Figure 1: a smooth monotone-skew smile (SPXW 2025-03-10 DTE 1, no knot). BBF0 sits ~ 67 bps from the Dupire PDE across the band while PHL1 and GHLOW2 close to a few bps. The bps gap is not the objective in itself — it is the measure of how far each map is from the model’s true Dupire implied vol, i.e. of the static-arbitrage slack PHL1 removes relative to BBF0.

Figure 2: a concave converted smile (DTE 3). The approximation remains good where vol is positive; the deep wings, where the option’s time value underflows and implied vol is not recoverable, are trimmed rather than faked.

Figure 3: the same DTE 1 cubic with a fake ATM knot (δ the real first-knot jump, moved to $k = 0$). PHL1 is visibly biased near the knot; PHL1 plus Φ_{BB}^{dir} tracks the PDE.

6 Conclusion

BBF calibrates any smile but gives no arbitrage guarantee; PHL1 supplies that guarantee but a single smooth cubic through it is too rigid to span the smiles BBF can reach. The obvious way to regain flexibility — a knot — was blocked by one defect: a piecewise cubic is only C^2 at a knot while PHL1’s σ_1 expansion assumes C^3 , so PHL1 develops a localised residual at the knot strike (BBF, a harmonic-mean integral, is insensitive to it and has no knot-specific error). This paper removed that single defect for a knot pinned at-the-money, in closed form: a one-dimensional Brownian-bridge integral $\Phi_{BB}(x, 0)$ with peak $3\sqrt{2\pi}/128$ and no fitted constants, which after subtracting the variation PHL1 already captures decays on both sides and adds directly to PHL1. On parameters calibrated from real SPXW surfaces it tracks the Dupire PDE where PHL1 alone is biased, and is inert on smooth smiles.

The consequence is practical, not just numerical: “piecewise-cubic with an at-the-money knot, mapped through PHL1” is now a usable surface parametrisation — it keeps PHL1’s arbitrage reassurance while recovering the calibration flexibility a single smooth cubic lacks, the knot adding shape freedom exactly where the smile is most curved. The at-the-money restriction is essential, not cosmetic: off-the-money the same construction leaves an unbounded polynomial requiring an ad-hoc truncation; $w = 0$ removes that term and fixes the peak by a Beta-function moment. The peak $3\sqrt{2\pi}/128$ is the C^2 companion, in log-normal Black IV, to the C^0 normal-IV constant $\frac{1}{16}\sqrt{\pi/2}$ of Costeanu and Pirjol — adjacent slots of the same Duhamel-perturbation table they began. Extending the bounded form to $w \neq 0$, and to the second-order $\Delta\gamma$ term, is the natural next step.

7 Reproducibility

All numbers and figures are produced by the open TypeScript core in the companion repository, cross-checked against an independent Python reference to 10^{-6} (closed forms) and 10^{-9} (the kernel). The figures are deterministic: regenerating them and diffing reproduces this paper exactly.

Credits. Market data for the calibrated example smiles: ThetaData.

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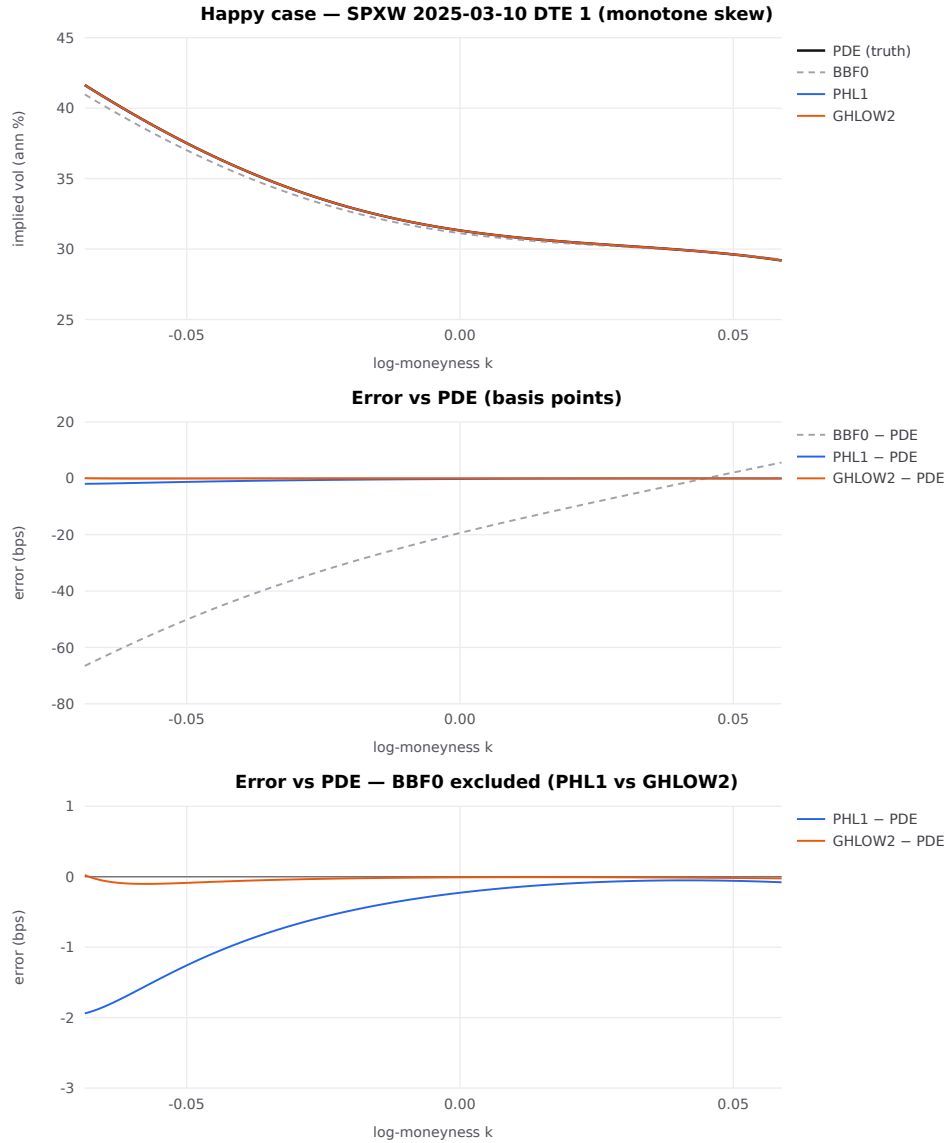


Figure 1: Happy case: smile (top), error vs PDE in bps with BBF0 (middle), and the same error with BBF0 excluded (bottom) so the PHL1-vs-GHLOW2 comparison is legible.

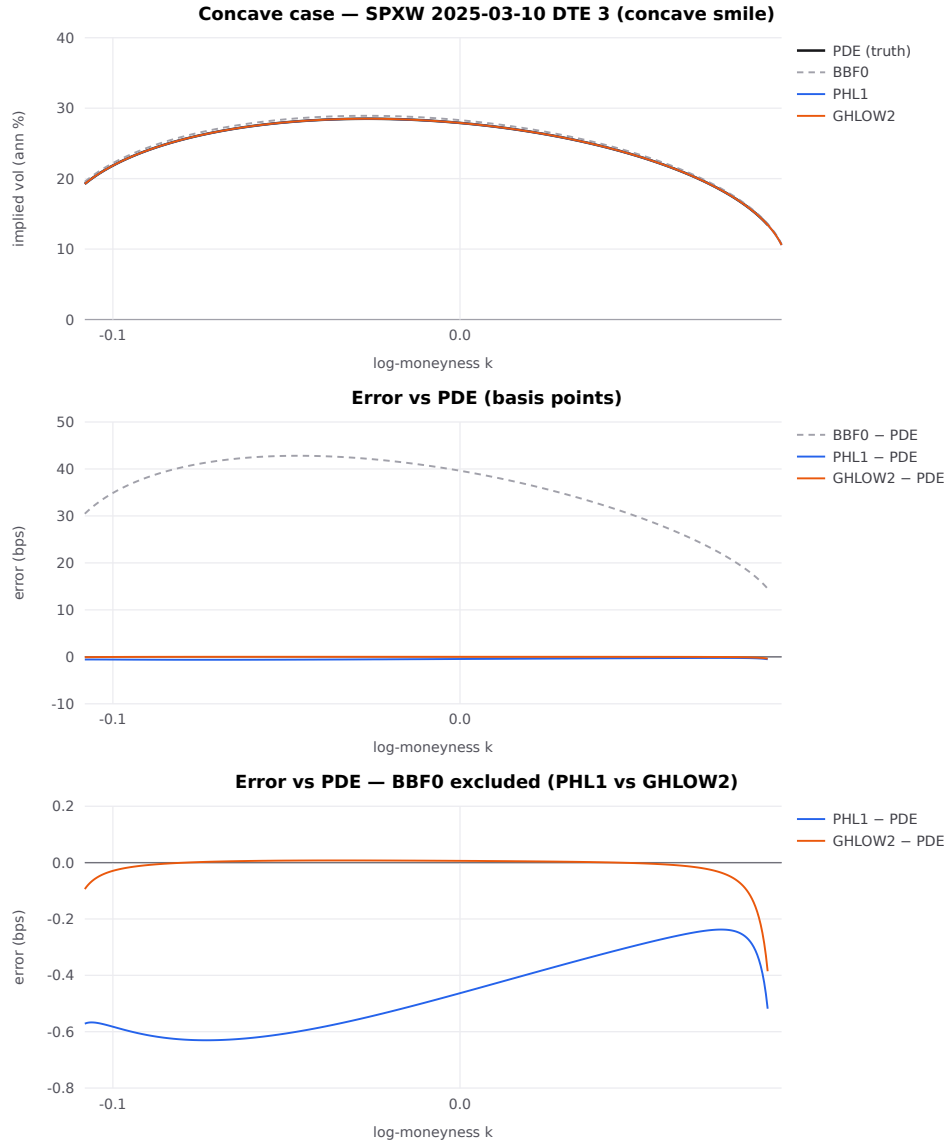


Figure 2: Concave case: the converted smile is concave. Panels as in Figure 1 (smile; error with BBF0; error without BBF0).

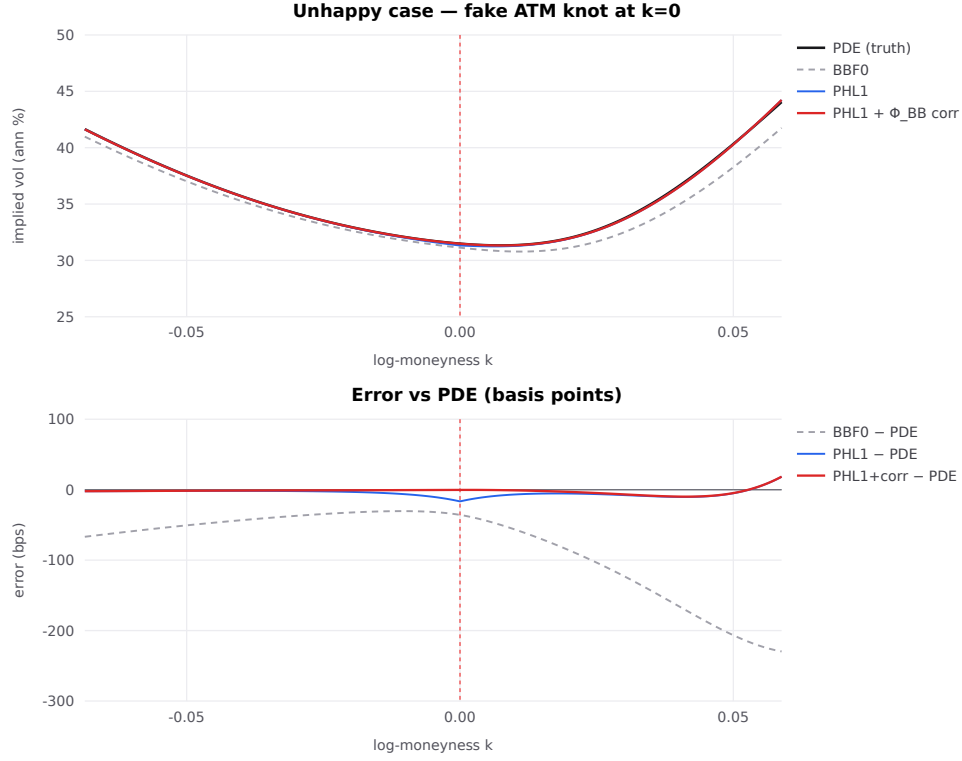


Figure 3: Unhappy case: a fake at-the-money knot. PHL1+correction recovers the PDE that PHL1 alone misses near $k = 0$.

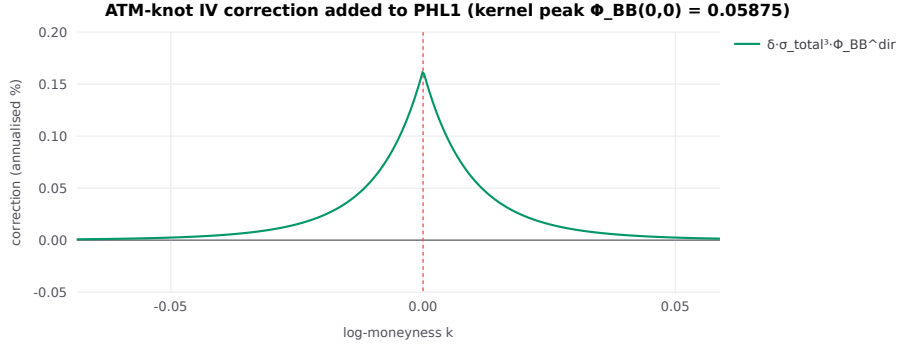


Figure 4: The applied ATM-knot correction $\delta \sigma_{\text{total}}^3 \Phi_{BB}^{\text{dir}}(k/\sigma_{\text{total}}, 0)$ in annualised % for the knot case (exactly the gap added to PHL1 in Figure 3). It is the dimensionless directed kernel — peak $\Phi_{BB}(0, 0) = 3\sqrt{2\pi}/128$, decaying on both sides after the PHL1 subtraction — scaled by $\delta \sigma_{\text{total}}^3$.